

Burnout in the pharmaceutical activity: The impact of COVID-19

Pedro Machado dos Santos (1,2), Claudia Ribeiro da Silva (5), Diana Costa (3), Carla Torre (3,4)

¹ CINTESIS - Center for Health Technology and Services Research

² Instituto de Ciências Biomédicas Abel Salazar da Universidade do Porto

³ Ordem dos Farmacêuticos

⁴ Faculdade de Farmácia da Universidade de Lisboa

⁵ Escola Superior de Saúde do Alcoitão

Corresponding author:

Name: **Pedro Machado dos Santos - P.M.S.**

Institutions: CINTESIS - Center for Health Technology and Services Research & Instituto de Ciências Biomédicas Abel Salazar da Universidade do Porto

Postal address: Rua Bartolomeu Dias, N.º84, 5º 1400-029 Lisboa, Portugal

Email: pedromacsantos@gmail.com

Telephone: +351 919442355

Co-authors

Name: **Cláudia Ribeiro da Silva - C.R.S.**

Institution: Escola Superior de Saúde do Alcoitão

City: Lisbon

Country: Portugal

Name: **Diana Costa - D.C.**

Ordem dos Farmacêuticos

Institution: Ordem dos Farmacêuticos

City: Lisbon

Country: Portugal

Name: **Carla Torre - C.T.**

Ordem dos Farmacêuticos

Institution: Ordem dos Farmacêuticos & Faculdade de Farmácia da Universidade de Lisboa

City: Lisbon

Country: Portugal

Highlights	3
Abstract	3
Strengths and limitations of this study	4
Strengths	4

Limitation	4
Keywords	4
Introduction	4
Methods	6
Study design	6
Patient and Public Involvement	6
Administration of Online Survey	6
Ethical Approval	7
Data collection and Inclusion Criterion	7
Outcome Measures	7
Burnout Survey Instrument	7
Statistical Analysis Plan	9
Results	9
Normative Data	10
Dependent variable Burnout	14
Dependent variable Emotional Exhaustion (EE)	14
Dependent variable Depersonalization (DP)	14
Dependent variable (Low) Personal Accomplishment (PA)	15
Discussion	17
Conclusion	21
Funding	22
Acknowledgments	22
Declaration of competing interest	22
Contributorship statement	22
Data sharing statement	22
References	22

Highlights

- This research identified five different profiles on the continuum between “burnout” (the most negative experience) to “engagement” (the most positive), based on a person-centred analysis of pharmacist’s burnout scores.
- Each profile has distinct relationships with various work life factors, which suggests that each profile reflects a different work life crisis that would require a unique intervention strategy.

- Having more confidence as a health care professional, on-the-job training, and direct patient care interactions may account as a buffering effect on burnout during Covid-19 pandemic.

Abstract

Objectives: Pharmacists are among the healthcare professionals involved in the response to COVID-19 pandemic, maintaining essential services. The study aimed to assess the occurrence of pharmacist's burnout and determine outcomes for each of its dimensions; identify potentially associated characteristics; and determine profiles and critical limits.

Design and setting: Cross-sectional observational study comprising an electronic survey addressed to Portuguese Pharmaceutical Society members.

Participants: From a total of 1362 participants, 91.4% were involved in direct and 7.7 % in non-direct patient care activity.

Outcome measures: The statistical analysis, performed with 2-tailed significance set at $P < .05$, yielded profiles based on the normalized (z) values for the MBI dimensions. Logistic regression was performed to identify the risk factors associated with burnout, and χ^2 test was used to compare the categorical variables training in the COVID-19 area with the confidence in caring for patients with this disease.

Results: Concern for the future (77.25%), workload (72.7%), scarcity of resources (62.4%), and loss of control over the activity (54.6%) were the main influencing factors on work-related feelings. Pharmacists who had been trained in COVID-19 were the most confident and showed a higher efficacy/outcome rate. Workload appears only as a predictor of high EE, with great impact on those who worked during the 3-month period more than 50 hours per week and on those who worked almost every weekend. The latter with greater predictive power of high EE ($p \leq 0.001$).

Conclusions: The new profile-based approach provides an understanding of the burnout experience and distinct relationships with various work-life factors and suggests that each profile reflects a different work-life crisis that would require a unique intervention strategy. Pharmacists who have been involved in direct care activities are at the highest risk of burnout. However, they were able to avoid depersonalisation and cynicism and thereby burnout syndrome.

Strengths and limitations of this study

Strengths

- This was the first study to look at the impact of COVID-19 on pharmacists across Portugal with a large sample size across all pharmacy sectors.
- This new focus on burnout profiles points to some new paths for future research and intervention on this important global problem.

Limitation

- Some non-direct patient care activity groups may have been disproportionately represented with larger numbers of responses from some contexts, for example, community and hospital pharmacists compared with regulatory affairs or pharmaceutical distribution.
- 85.5% of participants were women, potentially skewing results due to sample bias.

- The present results, indicating very positive developmental trends for work engagement and burnout, may not be applied to all occupational status groups.

Keywords

Burnout profiles; COVID-19; Maslach Burnout Inventory; Mental Health; Pharmacists; Pharmacy.

Introduction

Burnout Syndrome has been defined as a psychological disorder caused by excessive stress due to overload or overwork. In other words, it is a complex response to prolonged or chronic job stress [1, 2].

Despite appearing in the 10th revision of the World Health Organization (WHO) International Classification of Diseases (ICD) - chapter XXI: "factors influencing health status or contact with health services" - "burn-out" was only approved by the WHO as "occupational phenomenon", in May 2019 and will be officialised for the first time in the 11th revision of the ICD (ICD-11), scheduled for the year 2022 [2, 3].

In the ICD-11 proposal, this syndrome "results from chronic stress in the workplace that has not been well managed". It refers specifically to phenomena in the professional context and should not be applied to describe experiences in other areas of life. In describing "burn-out" the WHO contemplates the presence of 3 dimensions: (i) feelings of energy depletion or exhaustion; (ii) increased mental distance from one's job, or feelings of negativism or cynicism related to one's job; and (iii) reduced professional efficacy [2].

Although it is possible to find many evidence on burnout in healthcare professionals, most studies are focused on nurses and physicians [4, 5] and less on pharmacist profession [6]. According to Hall and colleagues, nurses had the greatest representation in the systematic review of 27 studies, followed by physicians, whereas pharmacists were represented in only 2 studies [4].

Like other healthcare professionals, pharmacists face many of the same challenges other providers do and are also prone to burnout due to several common and specific professional factors. This is also considering a highly regulated profession, with a continued focus on improving safety, therapeutic outcomes, and patient quality of life [7].

Despite the lack of research specific for pharmacists, evidence emerged showing that these professionals also suffered from burnout symptoms.

Moreover, the world, the healthcare sector and the pharmaceutical context in particular have undergone a huge change with a great influence on the delivery of pharmaceutical services. On March 11, 2020, the WHO declared the novel Coronavirus Disease 2019 (COVID-19) caused by the Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2) a pandemic [8]. The COVID-19 has severely affected the delivery of healthcare services worldwide and brought further pressures, causing additional stress and increased workload for pharmacy teams.

It is plausible that the COVID-19 pandemic may have exacerbated pre-existing causes of burnout by adding its own stressors. As healthcare workers, pharmacists played a role in hindering the spread of coronavirus by increasing patient awareness, especially by advising them on precautionary measures and providing appropriate information, and they were sometimes the source to provide protection products such as surgical masks or alcohol-based hand rub solutions. Community pharmacists were one of the most accessible healthcare professionals during this public health crisis, but all professional sectors have been functioning in a delicate balance of supporting users and families, colleagues, loved ones, and themselves.

In the national context of restrictions and reorganization of human resources, as a result of policies recently applied to the health sector, following international guidelines, and given the scarcity of data on burnout in pharmaceutical activity (mostly reported in academics studies), it was considered highly relevant to promote a nationwide survey aiming to collect more complete evidence on burnout syndrome among pharmacists and to understand how they have viewed their work and the people they worked closely with over the past 3 months and, by comparison, in the months prior to the pandemic (i.e. normative periods).

Therefore, the primary objective of this study was to determine pharmacists' outcomes for each of the 3 burnout dimensions: emotional exhaustion (EE), cynicism (CY) / depersonalization (DP), and personal accomplishment (PA). Secondary objectives included (i) an overall assessment of pharmacists burnout (ii) to identify potential characteristics associated with burnout (iii) to determine profiles and critical limits (cut-off points) for each of the 3 burnout dimensions.

Methods

Study design

The "Burnout Pharmaceutical Survey" was conducted by the Portuguese Pharmaceutical Society (PPS) with the support of the Center for Health Technology and Services Research (CINTESIS).

An advisory group of pharmacists, including academics and representatives of the PPS was set up to co-design the study, to develop the survey questions and to analyse the results. This multidisciplinary panel focused primarily on recent surveys about the impact of COVID-19 pandemic in health care activity and previous healthcare professional burnout studies published in peer-reviewed journals.

The questionnaire was designed for digital voluntary, confidential and anonymous participation. In total, 20 sections of multiple-choice questions were selected by the authors, favouring a more quick and spontaneous answers. Survey questions were divided into four segments of data collection: (i) demographics, (ii) employment and workplace characterisation, (iii) pandemic impact on labour activity and (iv) Burnout assessment (as described ahead).

Consensual review and validation were performed using a panel of experts who assessed and compared the different questions, according to the main board issues, semantics, idiomatic and conceptual equivalent of the items' contents. If determined as not applicable then it was modified to be more applicable, either to health care providers (pharmacists' patient care services) or non-providers. If there was no consensus, the majority of five panel member ruled on any issue.

Patient and Public Involvement

Two pharmacists were involved as research team members.

Professionals from community and hospital pharmacies who were at the forefront of the COVID-19 response were involved in (i) confirming the need and pertinence for conducting this research, (ii) identifying the main factors leading to pharmaceutical emotional distress ("burnout") and (iii) disseminating the survey.

A summary of results will be co-produced with pharmacists and disseminated to pharmacy professionals through PPS, International Pharmaceutical Society (FIP) and other representative organisations.

Administration of Online Survey

Google Forms was used to setup the survey online and to assemble responses, since it automatically hosts the online questionnaire via a unique URL and has proven to be an efficient and cost-effective platform to administer questionnaires without sacrificing quality, security, and fidelity of data [9]. This web-based and user-friendly platform was designed to support data capture for surveys and research studies, providing: (i) an intuitive interface for validated data entry; (ii)

automated export procedures for seamless data downloads to common statistical packages; and (iii) procedures for importing data from external sources. As recommended an account was solely created to manage the questionnaire data for the research team.

A pretest was performed with a convenience sample of twenty pharmacists, who reported that there were no issues with the contents of the survey. The final version was uploaded online using the PPS website.

Ethical Approval

The institutional review board of the of the Portuguese Pharmaceutical Society approved the project as exempt and without ethical reservations to its resolution, since (i) the procedures concerning the anonymity of the data and the information provided to the volunteers are foreseen and (ii) the nature of the elements to be collected does not carry the risk of revealing weaknesses unknown to the volunteer and thus triggering unpredictable reactions.

Data collection and Inclusion Criterion

The online survey was distributed via e-mail to all members of the PPS. To boost the potential for response, the questionnaire was disseminated through several organisations and networks including pharmaceutical sectorial and professional associations. It was also posted on social networks. The survey was cascaded by providing details of the study, soliciting participation in the survey and inviting pharmacists to share it with their professional networks and colleagues.

Selecting the electronic link served as first consent to participate voluntarily in the study. Information on the front page also confirmed that the survey was strictly confidential and in accordance with the applicable data protection law. Nevertheless, before starting the survey, study participants confirmed through a mandatory selection box that this was the first time completing this survey (ensuring 100% consent rate and preventing multiple responses) and acknowledged (i) the ethical terms, and (ii) the inclusion criteria as being a certified member of the PPS. No personal identifiers were linked to survey results to ensure privacy and the data, used only for research purposes, were maintained by the research team on a password-protected link. Contact e-mail from PPS was included for additional information.

Survey data were collected during a period of one month, from June 24 to July 24, 2020.

Outcome Measures

As in other studies assessing healthcare providers burnout, demographic and workplace characteristics collected in the survey included age, gender, number of people living in the household, years of practice, number of hours and weekends worked in the past 3 months (pandemic period) and degree of confidence in providing pharmaceutical care to patients with COVID-19.

Burnout Survey Instrument

Burnout was measured using the Maslach Burnout Inventory [10], currently recognised as a gold standard tool (reliable, valid and easy to administer) and the leading instrument to assess burnout [11].

The Portuguese versions of the MBI - HSS (Human Services Survey) e GS (General Services) - were used in this study after authorisation from the copyright holder of the instrument [12].

The 22 standardized questions from the MBI-HSS and the 15 questions from the MBI-GS reported the feelings at work, to be answered by means of a seven-point frequency scale ranging from 0 (never) to 6 (always). The instrument assesses the 3 major dimensions or constructs of burnout syndrome: emotional exhaustion (EE), depersonalization (DP) / cynicism (CY) and lack of personal accomplishment (PA), with each aspect measured by a separate scale. For EE and DP/CY subscales, high average values correspond to high levels of burnout symptoms. Low average values in the PA subscale

correspond to high levels of burnout symptoms. Although independent from the other two, PA cannot be seen as an opposition [10].

In the literature, it is possible to find a wide range of score cutoffs for burnout based on MBI results [13]. Thus, for the primary analysis of determining the proportion of pharmacists defined as burned out, we used high and low cutoff scores for the MBI subsets as defined by the original study authors. When stratifying burnout based on each subset score, we used the definition of burnout on the MBI scale of a high EE score of ≥ 27 , a high DP score of ≥ 10 , and/or low PA score of ≤ 33.17 . These cutoffs have been used most frequently in the health care professional literature.

For a more rigorous outcome measures and aiming to do a better risk assessment, the most recent guidelines were followed [14]. These contribute to identify different clinical profiles (Table 1), combining the results of the 3 sub-scales (EE, DP and PA) and based on "cut-off points" adjusted to the population under study.

Table 1 - Clinical Profiles based on the results of the burnout assessment

Profile	Description	Example
Engaged	Emotional stability, empathy, and professional fulfillment.	Professional with the ability to regulate himself emotionally, to empathize with others, and with feelings of professional effectiveness.
Ineffective	Low effectiveness/achievement rate (no exhaustion or depersonalization/cynicism problems).	Professional that reflects a loss of confidence in one's abilities, perhaps as a result of seemingly routine work or an environment that offers little recognition for a job well done.
Overextended	More acute form related to work overload (exhaustion).	Professional who is dedicated to his work and has a strong sense of efficiency, but who feels exhausted due to long working hours - he is accomplished and involved, but very tired.
Disengaged	Disconnection with the organization, its members, culture or values - high depersonalization/cynicism score (not overwhelmed)	Professional who demonstrates problematic relationships with co-workers and/or organizational values.
Burnout	Feeling of exhaustion, increased mental distance/feelings of work-related negativism or cynicism, and a sense of lack of professional/personal accomplishment.	A state of emotional exhaustion, with disbelief in the usefulness of the job one does and low achievement, with a decrease in effectiveness due to lack of commitment, which can lead to leaving the profession.

The results were analyzed separately, taking into account the pharmaceutical activity: direct patient care and non-direct patient care. The former, including the professional areas of (i) Community Pharmacy, (i) Hospital Pharmacy, and (iii) Laboratory Medicine and/or Human Genetics, were assessed with MBI-HSS. The latter composed of (iv) Regulatory Affairs, (v) Pharmaceutical Distribution, (vi) Academia, (vii) Pharmaceutical Industry and Scientific Research were assessed with the MBI-GS.

Statistical Analysis Plan

Data were fed into Epi Info and exported to SPSS statistical software for analysis.

The EpiInfo software, developed by the Centers for Disease Control and Prevention, U.S. Department of Health and Human Services [14], allowed us to ascertain that in a population of 16219 pharmacists (members of the PPS), the minimum recommended sample size, with a margin of error of 5% and a confidence interval of 99.9%, should be 1015 individuals.

The same formulas suggested by Leiter & Maslach, [15] were used to determine the critical limits for each of the 3 dimensions of the MBI. These calculations were performed for the sample of subjects in direct patient care activity using the MBI-HSS scale (22 items) and for the sample of subjects who were not in care activity using the MBI-GS scale (16

items). In their study the authors defined profiles using standardized (z) values for the sample [15]. Specifically, it was defined the critical threshold for (i) EE at $z = \text{Mean} + (\text{SD} * .5)$; (ii) DP/CY at $z = M + (\text{SD} * 1.25)$; and (ii) PA at $z = M + (\text{SD} * .10)$ [15].

These critical thresholds thus vary with context and depend on population norms for the group. Therefore, the categorisation of an individual's profile may slightly differ according to the population used to calculate the critical threshold.

Based on the critical limits, it was possible to perform a frequency analysis to determine in both samples how many subjects are below or above these limits in each of the 3 dimensions and establish the different profiles by combining the results in these factors.

In order to understand which variables may be related to Burnout, a logistic regression model was used, since the dependent variable of burnout is dichotomous (Yes/No). The independent variables used were: length of activity, number of working hours in the last 3 months, number of weekends worked in the last 3 months, degree of confidence to provide pharmaceutical care to patients with COVID-19, training in the COVID-19 area, living alone, and gender. Following the same rationale, 3 more logistic regressions were performed for each of the dichotomized MBI dimensions, using the same independent variables, to determine which ones might be related to high EE and DP and low PA.

The Chi-square test was also used to relate the variable training in the COVID-19 area (trained vs untrained) with the confidence in caring for patients with this disease (not at all or not very confident vs very or very confident).

Results

A total of 1362 pharmacists participated in the study. Of these 91.4% (n=1 246) were involved in direct patient care activity and 7.7% (n=106) in non-direct patient care activity.

Most respondents were women (85.5%), worked in Lisbon (29.8%), full time (96%), in the context of Community Pharmacy (77%) and had the possibility of receiving overtime (51.7%). The most represented group has been working for less than 5 years (23.2%), is between 31 and 40 years old (29%), is married/cohabiting (57.8%), lives with another person (28.3%), no children under 12 (62.5%) and no older adults over 65 (83.8%). While in 2019, they had worked between 31 and 40 hours (54.3%) per week and an average of two weekends per month (41.6%), in the first 3 months of the COVID-19 pandemic (2020) it changed to working between 41 and 50 hours (39.8%) per week. Although working two weekends a month remained the most common situation, there was an increase in the number of people who worked every weekend and a decrease in those who did not work on weekends.

The main sources of information to support the provision of care to COVID-19 patients were the Health Authorities (88.8%), the PPS (64.3%) and Sector Associations (44.4%).

Also, 41.1% of Pharmacists were not confident in providing care to COVID-19 patients and were most concerned about (i) infecting the family (88.5%), (ii) becoming infected while practicing (65.1%) and (iii) well-being of family members (64.7%). Almost half of the respondents were concerned about mental health (41.5%), well-being of their colleagues (41.3%) and professional burnout (41.1%).

Professional activity was mostly influenced by worry about the future (89.6%), workload (82.9%) and incomplete information and uncertainty (66.8%). Concern about the future (77.2.5%), workload (72.7%), scarcity of resources (62.4%), and loss of control over the activity (54.6%) were the main influencing factors on work-related feelings.

In our study, we found that the most affected dimension among pharmacists in care activity was PA (50.1%) followed by EE (35.8%), and DP (12%), regardless of having changes in isolation or associated with another dimension. In the sample of subjects in non-care activity the most affected dimensions was PA (41.5%) followed by EE (33%), and CY (19.8%).

Normative Data

In the present study, the same formulas suggested by Leiter & Maslach [15] were used to determine the critical thresholds and gauge the different profiles. These calculations were performed for the sample of subjects in care (N=1 246) using the MBI-HSS scale (22 items) and for the sample of subjects who were not in care (N=106) using the MBI-GS scale (16 items).

The critical boundaries found in this study for the different dimensions of the MBI-GS and MBI-HSS were respectively: EE ≥ 3.83 and ≥ 3.76 ; CY/DP ≥ 4.11 and ≥ 2.83 ; and PA ≤ 4.99 and ≤ 3.56 (Table 2).

Table 2 - MBI-GS and MBI-HSS scores to determine Burnout

Scale	Burnout dimensions	N	Mean	Std. Deviation	Ex + Sd*0,5	Dep + Sd*1,25	Real + Sd*0,1
MBI-GS	Emotional exhaustion	106	2.91	1.84	3.83		
	Cynicism	106	2.01	1.68		4.11	
	Personal accomplishment	106	4.89	0.98			4.99
MBI-HSS	Emotional exhaustion	1246	2.97	1.59	3.76		
	Depersonalization	1246	1.3	1.23		2.83	
	Personal accomplishment	1246	3.46	0.98			3.56

In the sample of subjects who were involved in direct care activities, we found a total of 7.3% of subjects in burnout (with the 3 dimensions of the MBI-HSS affected). Without any affected dimension of the MBI, a total of 36.4% of subjects ("engaged") were observed. With one dimension affected and at risk of burnout, there were 37.2% of subjects, and with two dimensions affected and at high risk of burnout, 19.4%. The most frequent profiles observed in this sample, were the "engaged" profile (good results in the 3 dimensions) followed by the "ineffective" profile (25.6%), "overextended and ineffective" (15.3%) and "overextended" (10.7%) (Table 3). When observing the altered dimensions, it was found that the most affected, regardless of having alterations isolated or associated with another dimension, is PA (50.1%) followed by EE (35.8%), and lastly DP (12%) (Table 3 and Figure 1).

Among the sample of pharmacists who were not involved in direct care activities, we found 14.2% in a situation of "burnout" (with the 3 dimensions of the MBI-HSS affected), 11.2% with high risk of burnout (two dimensions affected), 29.3% at risk of burnout (one dimension affected) and 45.3% were involved/engaged (no dimension affected). As for the sample profiles, we observe that the most frequent is the "engagement" (45.3%) (good results in the 3 dimensions) followed by the "ineffective" profile (18.9%), and "burnout" (14.2%) (Table 3 and Figure 2). With respect to the analysis of the altered dimensions, we see that the most affected is PA (41.5%) regardless of having alterations alone or associated with another dimension, followed by EE (33%), and lastly cynicism (19.8%).

Figures 1 and 2 show the means of the 3 burnout dimensions for each profile, in the two main areas of activity: direct patient care activity and non-direct patient care activity.

Table 3 - Distribution of participants by clinical profile and burnout risk

	Caring Activity MBI-HSS (n=1246)	Non Caring Activity MBI-GS (n=106)
--	-------------------------------------	---------------------------------------

Engaged (normal)	▼Exhaustion +▼Depersonalisation +▲Efficacy	1	36.4%	1	45.3%
Ineffective	▼Personal Accomplishment	2	25.6%	2	18.9%
Overextended	▲Emotional Exhaustion	4	10.7%	4	8.5%
Disengaged	▲Depersonalisation/Cynicism	8	0.6%	7	1.9%
Low to moderate risk			36.9%		29.3%
Ineffective & Overextended	▲Emotional Exhaustion +▼Personal Accomplishment	3	15.3%	5	7.5%
Overextended & Disengaged	▲Emotional Exhaustion +▲Depersonalisation/Cynicism	6	2.5%	6	2.8%
Ineffective & Disengaged	▲Depersonalisation/Cynicism +▼Personal Accomplishment	7	1.6%	8	0.9%
High risk			19.4%		11.2%
Burnout	▲Exhaustion +▲Depersonalisation +▼Efficacy	5	7.3%	3	14.2%

Figure 1 - Pattern of MBI-HSS Subscales across profiles: pharmacists involved in direct patient care activity

Profiles	Emotional Exhaustion	Depersonalization	Personal Accomplishment
cut-off point	(≥3.76)	(≥2.83)	(≤3.56)
Engaged 36.4% (n=453)	Low 1.79	Low 0.64	High 4.33
Ineffective 25.6% (n=319)	Low 2.19	Low 0.90	Low 2.66
Overextended 10.7% (n=133)	High 4.50	Low 1.30	High 4.07
Disengaged 0.6% (n=8)	Low 2.79	High 3.45	High 4.12
Ineffective & Overextended 15.3% (n=191)	High 4.66	Low 1.56	Low 2.74
Overextended & Disengaged 2.5% (n=31)	High 4.77	High 3.92	High 3.95

Figure 1 - Pattern of MBI-HSS Subscales across profiles: pharmacists involved in direct patient care activity

Profiles	Emotional Exhaustion	Depersonalization	Personal Accomplishment
Ineffective & Disengaged 1.6% (n=20)	Low 2.98	High 3.34	Low 2.86
Burnout 7.3% (n=91)	High 5.17	High 3.95	Low 2.45
100% (n=1,246)	Severe Low to moderate Normal		

Figure 2 - Pattern of MBI-GS Subscales across profiles: pharmacists with non-direct patient care activity.

Profiles	Emotional Exhaustion	Depersonalization	Personal Accomplishment
cut-off point	(≥3.83)	(≥4.11)	(≤4.99)
Engaged 45.3% (n=48)	Low 1.63	Low 1.05	High 5.49
Ineffective 18.9% (n=20)	Low 2.17	Low 1.26	Low 4.29
Overextended 8.5% (n=9)	High 5.18	Low 2.20	High 5.50
Disengaged 1.9% (n=2)	Low 2.70	High 4.20	High 5.50
Ineffective & Overextended 7.5% (n=8)	High 4.75	Low 2.10	Low 4.12

Figure 2 - Pattern of MBI-GS Subscales across profiles: pharmacists with non-direct patient care activity.

Profiles	Emotional Exhaustion	Depersonalization	Personal Accomplishment
Overextended & Disengaged 2.8% (n=3)	High 5.07	High 4.67	High 5.17
Ineffective & Disengaged 0.9% (n=1)	Low 3.40	High 6.00	Low 4.33
Burnout 14.2% (n=15)	High 5.29	High 4.87	Low 3.69
100% (n=106)			

■ Severe ■ Low to moderate ■ Normal

Dependent variable Burnout

The Omnibus test ($G^2(7)=74.868$, $p=0.000$) revealed that there was at least one independent variable in the model with predictive power over the dependent variable Burnout. The Hosmer & Lemeshow test revealed that the model is fitted ($X^2(8)$ HL=13.595 $p=0.093$). Through Wald's test we can see that the variables: length of activity ($X^2(1)=11.044$, $p=0.001$) degree of confidence to provide pharmaceutical care to patients with COVID-19 ($X^2(1)=26.739$, $p=0.000$), living alone ($X^2(1)=11.994$, $p=0.001$) and Gender ($X^2(1)=14.320$, $p=0.000$), showed a statistically significant result on the logit of the probability of having Burnout.

The probability of having Burnout was: (i) higher in those with less time of experience (1-5 years: 11.8%; 6-10 years: 9.3%; 11-15 years: 8.7%; 16-20 years: 4.5%; 21-25 years: 4.8%; 26-30 years: 1.1% burnout; 30 or more: 2.2%); (ii) higher in those who lack confidence in providing care to patients with COVID-19 (not at all confident: 20%; not very confident: 7.6%; fairly confident: 5.1%; very confident: 0.9%); (iii) higher in males (female: 6.5%; male: 12.3%); and (iv) higher in those who live alone (does not live alone: 6.3% of burnout; living alone: 14.8%).

The 3 logistic regressions performed had omnibus test results that revealed the existence of at least one independent variable in the model with predictive power over the dependent variables: EE, DP, (Low) PA. The Hosmer & Lemeshow test revealed that the model is fitted in all 3 regressions (Table 4). The results for each of the dependent variables are presented below.

Dependent variable Emotional Exhaustion (EE)

Through Wald's test we can see that the variables: number of hours worked in the last 3 months ($X^2(1) = 4.528$, $p = 0.03$), number of weekends worked in the last 3 months ($X^2(1) = 16.704$, $p = 0.000$), degree of confidence to provide pharmaceutical care to patients with COVID-19 ($X^2(1) = 14.618$, $p = 0.000$) and living alone ($X^2(1) = 10.560$, $p = 0.001$) showed a statistically significant result on the Logit of the probability of having EE.

The probability of having EE is: (i) higher in those who worked more hours in the last 3 months (20-30 hours: 23.5%, 31-40 hours: 35.9%, 41-50 hours: 34.4%, more than 50 hours: 44.1%); (ii) higher in those who have worked more weekends in the last 3 months (no weekends: 28.3%, 1-6 weekends: 28.9%, 7-10 weekends: 44.2%, all or almost all weekends: 45.4%); (iii) higher in those who lack confidence to provide care for patients with COVID-19: (not at all confident: 54.1%, not very confident: 36.5%, fairly confident: 30.6%, very confident: 35.8%); and (iv) higher in those who live alone (does not live alone: 34.3%; living alone: 47.2%)

Dependent variable Depersonalization (DP)

Through Wald's test we can observe that the variables: time in business ($X^2(1)=10.233$, $p=0.001$), degree of confidence to provide pharmaceutical care to patients with COVID-19 ($X^2(1)=20.673$, $p=0.000$), living alone ($X^2(1)=10.916$, $p=0.001$) and gender ($X^2(1)=12.103$, $p=0.001$) showed a statistically significant result on the Logit of the probability of having DP.

The probability of DP is: (i) higher in those who have less time of experience (1-5 years: 16%, 6-10 years: 17.4%, 11-15 years: 13.3%, 16-20 years: 9%, 21-25 years: 7.8%, 26-30 years: 3.2% and 30 or more: 6.7%); (ii) higher in those who are not confident to provide care to patients with COVID-19 (not at all confident: 23%, not very confident: 14.1%, fairly confident: 8.9%, very confident: 5.5%); (iii) higher in males (female: 11.1%, male: 17.9%) and (iv) higher in those who live alone (does not live alone: 11.0%; Living alone: 20.4%).

Dependent variable (Low) Personal Accomplishment (PA)

Through Wald's test we can observe that the variables: time in business ($X^2(1)=10.620$, $p=0.001$), degree of confidence to provide pharmaceutical care to patients with COVID-19 ($X^2(1)=31.562$, $p=0.000$), completed training in COVID-19 area ($X^2(1)=7.507$, $p=0.006$) and gender ($X^2(1)=5.885$, $p=0.015$) showed a statistically significant result on the Logit of the probability of having a low PA.

The probability of having a low personal accomplishment is: (i) higher in those with less time of experience (1-5 years: 57.3%, 6-10 years: 53% low achievement, 11-15 years: 53.8% low achievement, 16-20 years: 43% low achievement, 21-25 years: 42.2%, 26-30 years: 45.2%, 30 or more: 44.4%); (ii) higher in those who lack confidence in providing care to patients with COVID-19 (not at all confident: 68.1%, not very confident: 53.7%, fairly confident: 45.3%, very confident: 31.2%); (iii) higher in those who did not take COVID-19 training (no training: 54.8%; took training: 43.5%); and (iv) higher in males (male 55.3%; female: 48.9%).

Table 4 - Variables with predictive power over MBI-HSS scores (logistic regression)

					95% C.I. for EXP (B)					
Independent variables					B	S.E.	Wald	Exp (B)	Lower	Upper
DV: Burnout Omnibus: G(7)=74.868 p= 0.000 Hosmer and Lemeshow : X² (8)_{HI}=13.595 p=0.093	How long (in years) have you been working as a pharmacist?				-0,248	0,075	11,044***	0,780	0,674	0,903
	On average, how many hours per week have you worked in the past 3 months?				-0,059	0,126	0,220	0,943	0,736	1,207
	How many weekends have you worked in the past 3 months?				0,094	0,063	2,232	1,099	0,971	1,243
	How would you describe your degree of confidence in providing pharmaceutical care to patients with COVID-19?				-0,808	0,156	26,739***	0,446	0,328	0,605
	Have you completed any COVID-19 related training?				0,232	0,244	0,904	1,261	0,782	2,033

	Living alone	-0,997	0,288	11,994***	0,369	0,210	0,649
	Male	1,072	0,283	14,320***	2,920	1,676	5,086
	Constant	0,285	0,588	0,235	1,330		
DV: Emotional exhaustion Omnibus: $X^2=58.395$ $p=0.000$ Hosmer and Lemeshow : $X^2=10.826$ $p=0.212$	How long (in years) have you been working as a pharmacist?	-0,049	0,035	1,968	0,952	0,888	1,020
	On average, how many hours per week have you worked in the past 3 months?	0,140	0,066	4,528*	1,150	1,011	1,308
	How many weekends have you worked in the past 3 months?	0,136	0,033	16,704***	1,145	1,073	1,222
	How would you describe your degree of confidence in providing pharmaceutical care to patients with COVID-19?	-0,306	0,080	14,618***	0,737	0,630	0,862
	Have you completed any COVID-19 related training?	-0,058	0,128	0,206	0,943	0,733	1,213
	Living alone	-0,630	0,194	10,560***	0,532	0,364	0,779
	Male	0,319	0,177	3,245	1,376	0,972	1,948
	Constant	0,022	0,337	0,004	1,022		
DV: Depersonalization Omnibus: $X^2=64.568$ $p=0.000$ Hosmer and Lemeshow : $X^2=15.024$ $p=0.059$	How long (in years) have you been working as a pharmacist?	-0,176	0,055	10,233***	0,838	0,752	0,934
	On average, how many hours per week have you worked in the past 3 months?	-0,034	0,098	0,124	0,966	0,798	1,170
	How many weekends have you worked in the past 3 months?	0,045	0,049	0,833	1,046	0,950	1,151
	How would you describe your degree of confidence in providing pharmaceutical care to patients with COVID-19?	-0,543	0,119	20,673***	0,581	0,460	0,734
	Have you completed any COVID-19 related training?	0,015	0,188	0,006	1,015	0,702	1,468
	Living alone	-0,801	0,242	10,916***	0,449	0,279	0,722
	Male	0,807	0,232	12,103***	2,241	1,422	3,532
	Constant	0,292	0,471	0,383	1,339		
DV: Personal accomplishment Omnibus: $X^2=69.351$ $p=0.000$ Hosmer and Lemeshow : $X^2=9.096$ $p=0.334$	How long (in years) have you been working as a pharmacist?	-0,111	0,034	10,650***	0,895	0,837	0,957
	On average, how many hours per week have you worked in the past 3 months?	0,015	0,064	0,052	1,015	0,895	1,150
	How many weekends have you worked in the past 3 months?	0,048	0,049	0,941	1,049	0,952	1,155
	How would you describe your degree of confidence in providing pharmaceutical care to patients with COVID-19?	-0,439	0,078	31,562***	0,645	0,553	0,751
	Have you completed any COVID-19 related training?	0,339	0,124	7,507**	1,403	1,101	1,789
	Living alone	-0,164	0,195	0,707	0,849	0,580	1,243
	Male	0,427	0,176	5,885*	1,532	1,085	2,162
	Constant	1,542	0,338	20,839	4,673		

* Statistically significant to $p \leq 0,05$, ** Statistically significant to $p \leq 0,01$, *** Statistically significant to $p \leq 0,001$

Since the variable degree of confidence in providing pharmaceutical care to patients with COVID-19 was shown in all regressions to be predictive of burnout and of the dimensions involved in burnout, we investigated whether those who had been trained in COVID-19 were the most confident in caring for patients with this disease.

The Chi-square test revealed the existence of a significant association ($p \leq 0.001$) between training in COVID-19 and confidence to provide care to /patients with this disease: in the group of pharmacists who have taken training there was a higher percentage with quite or a lot of confidence to provide pharmaceutical care to patients with COVID-19 (50.5%), while this percentage is significantly lower (only 38.7%) in the group that has not taken training.

Discussion

Our study, with 1362 participants, far exceeded the recommended minimum sample size of 1015 subjects and represents one of the largest in the scope of burnout studies among pharmacists, at a global level. According with a systematic review on pharmacist burnout that addressed studies about US pharmacists practicing in any setting, through February 13, 2019, the largest sample was obtained in a study published in 1990 with 1258 professionals also working in multiple pharmacy practicing settings [16].

So far, to the best of our knowledge, it is the most representative sample regarding studies conducted with members enrolled in health professional associations in Portugal - e.g. Medical Society and Nurses Society [5].

Based on a person-centered analysis of people's burnout scores conducted by Leiter & Maslach [15] this research also identified five different profiles on the continuum between "Burnout" (the most negative experience) to "Engagement" (the most positive). Although we started attending the same three intermediate profiles - "Disengaged" (characterized by high DP/CY only), "Overextended" (high EE only), and "Ineffective" (low PA only) - we think suggest? that the 7-class model fit best to identify multiple person-centered forms of distress and profiles across the burnout.

A person-centered approach considers the whole person, in contrast with an exclusive concern on single dimensions prioritized in variable-centered approaches, and can be useful for theory, research, and designing interventions.

The Burnout profile is clearly more distressing than "overextended", but both have issues to be addressed. In the middle we find a "disengaged" profile more negative than "overextended", and closer to the "Burnout" profile, which argues against the use of exhaustion alone as a proxy for burnout.

Standardized (z) values used to calculate profiles with the critical thresholds among pharmacists confirmed that these critical boundaries vary across the context and are dependent on the population norms for the group. In this study it is possible to advance a new pharmacist's profile categorisation, based on the population used in the critical boundary calculation (direct patient care and non-direct patient care) and slightly different from those proposed by the authors [12].

Surprisingly, professionals who were involved in direct care activities, represented mainly by community pharmacists (77%), had a higher risk of burnout (19.4%) but effectively less burnout (with all 3 dimensions affected) compared to pharmacists who were not involved in direct care activities (7.3% and 14.2%, respectively).

The prevalence results are in line with an investigation conducted in the US with pharmacists in the health care system [17] where 53.2% of the study participants (n=177) showed scores indicating a high degree of burnout on at least one dimension and 8.5% respondents (n=28) scores indicating burnout on all 3 subscales of the (MBI-HSS: high EE, high DP, and reduced PA). According to the authors, this burnout rate was similar to published findings for physicians and nurses.

Regarding the associations between burnout and primary work setting, our results does not appear to support previous studies. An American earlier research revealed that those pharmacists working primarily in community chain store

settings reported higher levels of burnout than those working in hospital or institutional pharmacies, independent community pharmacies, academia, and home health care [18]. An English study found that community pharmacists had significantly higher levels of workplace stress than other healthcare professionals [19]. Most recently, a cross-sectional study also showed a high prevalence of psychological distress, work-related burnout and compassion fatigue among hospital pharmacists in Japan [6].

However, all the studies above mentioned were conducted before the COVID-19 pandemic situation. In this case, pharmacists who were on the front line (in direct care activities), despite the high increase in professional demands, maintained their routines and became less isolated, compared to those who were not involved in direct care activities and had to drastically change their daily lives. Community pharmacists, for instance, who were one of the most accessible healthcare professionals during this public health crisis, and here most represented (77%), were always in contact with the public, reinforcing human relationships that avoid depersonalisation and cynicism, and therefore burnout syndrome.

Nevertheless, our study results are framed in an early 2020 systematic review of burnout in pharmacists that revealed an overall estimate of the prevalence of burnout based on individual MBI subscales that ranged from 8% to 53% [20]. The prevalence of burnout reported in 6 articles (40%), also using specific scores to define MBI measures of high EE and low PA, showed a wider range and lower than previously reported by individual studies, which cited overall burnout ranging from 52% to 61% using a composite of one or more MBI subscales.

In contrast to this systematic review the proportion of pharmacists (involved in care and non-care activity) with MBI subscale scores consistent with burnout were greater for low PA (49.8% and 41.5%), than for high EE (35.8% and 33%). Yet, the percentage of subjects with high DP (12% and 19.8%) was always lower in the different studies.

The already mentioned meta-analysis showed different proportions in both groups of 9 and 6 studies reviewed, but still greater for high EE (41% and 37%) than for low PA (32% and 33%). The proportion of pharmacists who met authors' definition of high DP were always lower (20% and 19%) [20].

Our results are not in agreement with specific burnout studies among pharmacists conducted before the pandemic period, such as the research with hospital pharmacists in Romania, that identified a moderate level of EE, a low level for DP and an average level of burnout for PA [21]. In South Africa researchers reported that 39% of pharmacists in community pharmacies, hospital, and executive positions exhibited moderate to high levels of emotional exhaustion (EE) [22] and in Turkey community pharmacists revealed low levels of personal accomplishment (PA) [23]. And an American study, revealed that pharmacists practicing on the front lines of insurance structures (*health maintenance organization*) showed also moderate levels of EE and DP [24].

However, the recent "Latent Profile Analysis" [15] using a new data modeling approach and multiple person-centered profiles in 1766 health care employees, including a wide range of clinical, administrative, and support areas, has also highlighted "ineffective" (high inefficacy, moderate other) as the most frequent profiles (31%), immediately after "engagement" (44%). An order of prevalence which is in line with our study: "ineffective" 25.6% (MBI-HSS) and 18.9% (MBI-GS); and "engagement" 36.4% (MBI-HSS) and 45.3% (MBI-GS).

In all above-mentioned studies the hypotheses for the Ineffective profile were less clear. Perhaps because there has been less prior research on the efficacy dimension and because the existing findings do not show strong relationships between this dimension and the kind of workplace variables already assessed.

In our study, the difference on burnout subsets and an identifiable "high ineffective only" seems to be related to the current crises contexts: the onset of the pandemic period and the role played by pharmacists in hindering the spread of coronavirus. While being less confident to provide care to patients with COVID-19 showed a statistically significant probability of having a low PA, high EE, high DP and burnout as a whole, not take COVID-19 training has only statistically significant result on low accomplishment.

These findings not only suggest that ineffectiveness and loss of confidence in one's abilities are a consequence of labour insecurity feelings, but most of all these speaks to the importance of improving confidence among health professionals in a crisis context.

Since pharmacists who had received training in COVID-19 were the most confident, and showed a higher efficacy/outcome rate, it would appear clear that future contingency plans should include training for healthcare professionals, particularly those who have less work experience, creating a working environment that provides more recognition for a job well done, as suggested by Maslach Burnout Inventory authors [12].

Indeed, in our survey, the likelihood of having burnout, high PD and low PA is higher in those with less work experience, and probably less confident. This corroborates previous American research which revealed that pharmacists in their first 10 years of practice in all work settings had moderate levels of burnout [18].

More recent evidence has shown that the PA factor is significantly reduced in the "oldest" age category, but with an increasing length of practice the average of EE factor also increases [25]. The latter has not been yet confirmed in our study.

Workload in turn appears only as a predictor of high EE, with great impact on those who worked during the 3-month period more than 50 hours per week (44.1%) and on those who worked almost every weekend (45.4%). The latter with greater predictive power of high EE ($p \leq 0.001$). This not only underlines the importance of free time in an individual's wellbeing, but also recalls the evidence that weekends are associated with high feelings of freedom and closeness - engaging in activities of your choice and time with close friends and family [26] and that working extra-long hours and over weekends contributes to more depressive symptoms and worse mental health [27].

Outside a crisis, United States (US) researchers reported that two-thirds of pharmacists described their workload as high or excessively high, and 45% reported that their workload had negatively impacted their emotional and mental health [28].

In this recent and particular context, it is reasonable to expect that pharmacists living alone would have absorbed a greater workload and in a combined way led to a greater EE, DP and burnout syndrome. Even if this single condition has not affected PA, or on the contrary may improve this dimension, it is important to underline that isolation poses risks to mental health [29-31]. Consequently, it would be important to consider measures that promote regular contact with significant people in order to prevent depersonalisation and cynicism, and therefore burnout syndrome.

Until quite recently it was expected that environmental factors of the modern healthcare workplace would arguably continue to contribute not only to EE, but also to DP, and to reduced PA in the foreseeable future [32]. However, the knowledge acquired during the pandemic crisis may help professionals and decision-makers to break this trend and to clarify which individual and environmental weaknesses and strengths may contribute to explaining and coping better with burnout in pharmacists.

Despite the fact that pharmacists fall into virtually the same health care provider category, some pharmacy sectors has less information available (e.g. pharmaceutical industry). Our study follows the trend of representing community pharmacy more, in addition to relatively recent studies conducted among community pharmacists [19, 22, 23, 25, 33], but also at a pharmacy practice faculty [34] or among hospital pharmacists [6, 21]. Nevertheless, overall, these reports reflect an emerging picture of burnout that has existed, and continues to exist, in the pharmacy profession.

The evidence had already shown that EE among nurses and physicians working in intensive care units was associated with an increased of patient mortality [35] and a systematic analysis including providers from multiple healthcare professions showed an association between burnout and an increase of medical errors [4]. A more recent meta-analysis of 47 studies on 42 473 physicians described physician burnout as an epidemic affecting healthcare delivery and patient safety [36]. However, no pharmaceutical-specific papers were identified regarding the impact of burnout on job performance and patient safety.

It is important to note that the COVID-19 period covered is not an normal working situation. Therefore, it would be important to repeat the study after the pandemic, ideally collecting evidence about the association between pharmacists burnout and healthcare delivery and patient safety. Repeating the study at international level (during and after the pandemic situation) may also help to verify the validity of the cut-off points and reinforce the findings.

Conclusion

Burnout symptoms can affect pharmacists in all practice settings, yet most pharmacists do not meet the criteria for the individual dimensions of burnout. Inconsistencies on how this complex response to prolonged or chronic job stress has been measured in the literature limit conclusions and also recommendations for interventions.

This new profile-based approach provides an understanding of the burnout experience and distinct relationships with various work-life factors and suggests that each profile reflects a different work-life crisis that would require a unique intervention strategy. Therefore, it was extremely important to use this new methodology, determining the five MBI profiles for pharmacists and contributing to the identification of differentiated scores between healthcare professional groups.

Our data indicate that pharmacists who have been involved in direct care activities are at the highest risk of burnout compared with professionals practising in other areas. However, they were able to avoid depersonalisation and cynicism and thereby burnout syndrome.

The syndrome is often associated with issues surrounding workload, professional experience or degree of confidence in providing pharmaceutical care.

Prevention strategies such as focusing on work-life balance, peer support, continued education and self-care may be effective in reducing the risk of burnout among pharmacists. Individuals who demonstrate greater professional, psychological and social frailty, such as lack of experience, anxiety, depression or social isolation, require greater mental health attention and support.

Unfortunately, burnout is not a new challenge in many different settings and workplaces. A growing body of literature supports the notion that burnout symptoms are prevalent among healthcare providers and yet associated with negative outcomes, such as suboptimal patient care and professional inefficiencies.

It is therefore opportune to continue and deepen the knowledge about pharmacists mental health and its impact on people benefiting from pharmaceutical care and on co-workers, in order to allow better comparison between different professional groups and contribute to the adjustment of health policies.

In conclusion, the pandemic of COVID-19 has had a huge impact on the provision of pharmaceutical care. All efforts, logistical procedures and patient counselling were essential for maintaining high quality care and reducing harm, especially among the most vulnerable groups. In order to plan, provide better support to the pharmaceutical sector and prevent future crises, as well as to potentially adapt healthcare services to the cycles of SARS-CoV-2, interventions to promote psychological well-being of pharmacists would need to be developed.

Funding

This work was supported by the Portuguese Pharmaceutical Society. This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Acknowledgments

Authors would like to acknowledge the valuable comments to the questionnaire design from the President of the PPS, Ana Paula Martins, and the Members of the National Board, Helena Farinha and Ema Paulino, as well as all the pharmacists that contribute to this research.

Declaration of competing interest

None.

Contributorship statement

P.M.S., D.C. and C.T. conceived and planned the experiments. P.M.S., D.C. and C.T. carried out the experiments. C.R.S. and P.M.S. analysed the data. P.M.S., C.R.S., D.C. and C.T. contributed to the interpretation of the results. P.M.S. took the lead in writing the manuscript. All authors provided critical feedback and helped shape the research, analysis and manuscript.

Data sharing statement

The authors confirm that the data supporting the findings of this study are available within the article [and/or] its supplementary materials.

References

1. Maslach, C., & Leiter, M. P. (2007). Burnout. In Encyclopedia of Stress (pp. 368–371). Elsevier Inc. <https://doi.org/10.1016/B978-012373947-6.00062-3>
2. World Health Organization (2019). Burn-Out an “Occupational Phenomenon”: International Classification of Diseases; World Health Organization: Geneva, Switzerland; Available online: https://www.who.int/mental_health/evidence/burn-out/en/
3. World Health Organization (2020). International Statistical Classification of Diseases and Related Health Problems (ICD) ; World Health Organization: Geneva, Switzerland; Available online: <https://www.who.int/standards/classifications/classification-of-diseases>
4. Hall, L.H., Johnson, J., Watt, I., Tsipa, A., O'Connor, D.B. (2016). Healthcare Staff Wellbeing, Burnout, and Patient Safety: A Systematic Review. PLoS One. 11(7):e0159015.
5. Marôco, J., Marôco, A.L., Leite, E. et al. (2016). Burnout nos profissionais da saúde Portugal, Acta Med Port. Jan; 29(1):24-30
6. Higuchi Y, Inagaki M, Koyama T, et al. (2016). A cross-sectional study of psychological distress, burnout, and the associated risk factors in hospital pharmacists in Japan. *BMC Public Health*;16:1–8
7. Liekweg, A., Westfeld, M., Jaehde, U. (2004). From oncology pharmacy to pharmaceutical care: new contributions to multidisciplinary cancer care. *Support Care Cancer*. 12(2):73–9.

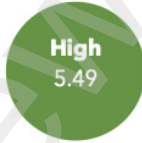
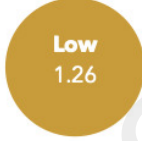
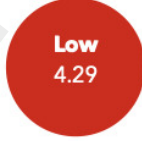
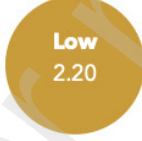
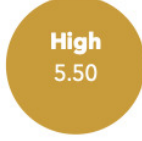
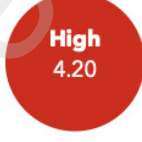
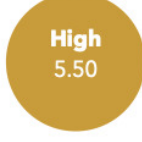
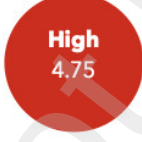
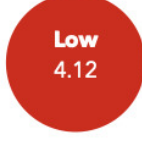
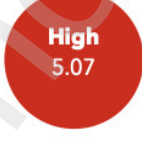
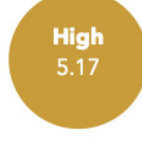
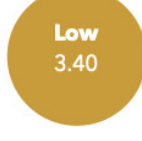
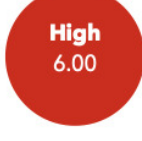
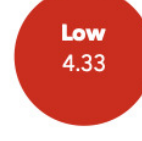
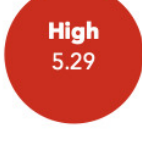
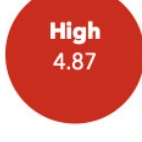
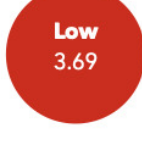
8. World Health Organization (2020). WHO director-general's opening remarks at the media briefing on COVID-19 - 11 March 2020. <https://www.who.int/dg/speeches/detail/who-director-general-s-opening-remarks-at-the-media-briefing-on-COVID-19---11-march-2020>
9. Rayhan, R. U., Zheng, Y., Uddin, E., et al. (2013). Administer and collect medical questionnaires with Google documents: a simple, safe, and free system. *Applied medical informatics*, 33(3), 12–21.
10. Maslach, C., Jackson, S.E., Leiter, M.P., et al. (1996). Maslach Burnout Inventory Manual. Palo Alto: University of California, Consulting Psychologist Press.
11. Boudreau, R.A.; Boudreau, W.F.; Mauthe-Kaddoura, A.J. (2015). From 57 for 57: A Bibliography of Burnout Citations. In Proceedings of the 17th Conference of the European Association of Work and Organizational Psychology (EAWOP): Oslo, Norway.
12. MindGarden (2019). A message from the Maslach Burnout Inventory Authors. Blog Post: March 19. Mind Garden, Inc.
13. Doulougeri, K., Georganta, K., Montgomery, A. (2016). “Diagnosing” burnout among healthcare professionals: Can we find consensus? *Cogent Med.* 3(1):1–10.
14. Dean, A., Arner, T., Sunki, G., Friedman, R., et al. (2007). Epi Info™, A database and statistics program for public health professionals. Centers for Disease Control and Prevention. Atlanta, Georgia, USA.
15. Leiter, M. P., & Maslach, C. (2016). Latent burnout profiles: A new approach to understanding the burnout experience. *Burnout Research*, 3(4), 89–100. <https://doi.org/10.1016/j.burn.2016.09.001>
16. Lahoz, M.R. & Mason, H.L. (1990). Burnout among pharmacists. *Am Pharm.* 30(8):28–32.
17. Durham, M.E., Bush, P.W., Ball, A.M. (2018). Evidence of burnout in health-system pharmacists. *Am J Health Syst Pharm.* 1;75(23 Supplement 4):S93-S100; 2018. doi: 10.2146/ajhp170818. Epub 2018 Oct 17.
18. Barnette, C.W., Hopkins, W.A. and Jackson, R.A. (1986). Burnout experienced by recent pharmacy graduates of Mercer University. *Am J Hosp Pharm.* 43:2780–4.
19. Jacobs, S., Hassell, K., Ashcroft, D., et al. (2014). Workplace stress in community pharmacies in England: associations with individual, organizational and job characteristics. *J Health Serv Res Policy.* 19:2733.
20. McQuade, B.M., Reed, B.N., Didomenico, R., et al. (2020). Feeling the burn? A systematic review of burnout in pharmacists. *Journal of the American College of Clinical Pharmacy*, 3, 663 - 675.
21. Iorga, M., Dondas, C., Sztankovszky, L.Z., Antofie, I. (2018) Burnout Syndrome among Hospital Pharmacists in Romania. *Farmacia.* 66, 181–188.
22. Rothmann, S., Malan, A.M. and Rothmann, J.C. (2001) Sense of coherence, coping, and burnout in a corporate pharmacy group. Paper presented at Seventh Annual Psychological Society of South Africa Conference. Johannesburg, South Africa; Aug.
23. Calgan, Z., Aslan, D. and Yegenoglu, S. (2011). Community pharmacists' burnout levels and related factors: an example from Turkey. *Int J Clin Pharm.* 33:92–100.
24. Gupchup, G.V., Singhal, P.K., Dole, E.J. and Lively, B.T. (1998). Burnout in a sample of HMO pharmacists using the Maslach Burnout Inventory. *J Manag Care Spec Pharm.* 4:495–503.
25. Saligerová, M. & Kolar, J. (2017). The extent of the burnout syndrome among pharmacists: partial study. *Ceska Slov Farm*;66:10711.

26. Ryan, R. M., Bernstein, J. H., & Brown, K. W. (2010). Weekends, work, and wellbeing: Psychological need satisfactions and day of the week effects on mood, vitality, and physical symptoms. *Journal of Social and Clinical Psychology*, 29, 95–122.
27. Weston, G., Zilanawala, A., Webb, E., Carvalho, L.A., McMunn, A. (2019). Long work hours, weekend working and depressive symptoms in men and women: findings from a UK population-based study. *J Epidemiol Community Health*. 73(5):465-474. doi: 10.1136/jech-2018-211309. Epub 2019 Feb 25. PMID: 30804048; PMCID: PMC6581113.
28. Midwest Pharmacy Workforce Research Consortium (2015). 2014 National Pharmacist Workforce Survey
29. Holt-Lunstad, J., Smith, T.B., Baker, M., et al. (2015). Loneliness and Social Isolation as Risk Factors for Mortality: A Meta-Analytic Review. *Perspectives on Psychological Science*. 2015;10(2):227-237. doi:10.1177/1745691614568352
30. Reger, M.A., Stanley, I.H., Joiner, T.E. (2020). Suicide mortality and coronavirus disease 2019—a perfect storm? *JAMA Psychiatry* (Epub April 10, 2020). 10.1001/jamapsychiatry.2020.1060
31. Young, K.P., Kolcz, D.L., O’Sullivan, D.M., Ferrand, J., Fried, J., Robinson, K. (2020). Health Care Workers' Mental Health and Quality of Life During COVID-19: Results From a Mid-Pandemic, National Survey. *Psychiatr Serv*. 72(2):122-128. doi: 10.1176/appi.ps.202000424. Epub 2020 Dec 3. PMID: 33267652.
32. Zellmer, W.A., (2016) ed. Pharmacy forecast 2017: strategic planning advice for pharmacy departments in hospitals and health systems. *Am J Health-Syst Pharm*. 73:e617–43.
33. Balayssac, D., Pereira, B., Virot, J., et al. (2017). Burnout, associated comorbidities and coping strategies in French community pharmacies—BOP study: a nationwide cross-sectional study. *PLoS One*; 12:e0182956.
34. El-Ibiary, S.Y., Yam, L. and Lee, K.C. (2017). Assessment of burnout and associated risk factors among pharmacy practice faculty in the United States. *Am J Pharm Educ*. 81: article 75
35. Welp, A., Meier, L.L. and Manser, T. (2015). Emotional exhaustion and workload predict clinician-rated and objective patient safety. *Front Psychol*.; 5:1573.
36. Panagioti, M., Geraghty, K., Johnson, J., et al. (2018). Association Between Physician Burnout and Patient Safety, Professionalism, and Patient Satisfaction. *JAMA Intern Med*. 178(10):1317-1330. doi:10.1001/jamainternmed.2018.3713.

Figure 1 - Pattern of MBI-HSS Subscales across profiles: pharmacists involved in direct patient care activity

Profiles	Emotional Exhaustion	Depersonalization	Personal Accomplishment
cut-off point	(≥ 3.76)	(≥ 2.83)	(≤ 3.56)
Engaged 36.4% (n=453)	Low 1.79	Low 0.64	High 4.33
Ineffective 25.6% (n=319)	Low 2.19	Low 0.90	Low 2.66
Overextended 10.7% (n=133)	High 4.50	Low 1.30	High 4.07
Disengaged 0.6% (n=8)	Low 2.79	High 3.45	High 4.12
Ineffective & Overextended 15.3% (n=191)	High 4.66	Low 1.56	Low 2.74
Overextended & Disengaged 2.5% (n=31)	High 4.77	High 3.92	High 3.95
Ineffective & Disengaged 1.6% (n=20)	Low 2.98	High 3.34	Low 2.86
Burnout 7.3% (n=91)	High 5.17	High 3.95	Low 2.45
100% (n=1,246)			
	Severe	Low to moderate	Normal

Figure 2 - Pattern of MBI-GS Subscales across profiles: pharmacists with non-direct patient care activity.

Profiles	Emotional Exhaustion	Depersonalization	Personal Accomplishment
cut-off point	(≥ 3.83)	(≥ 4.11)	(≤ 4.99)
Engaged 45.3% (n=48)			
Ineffective 18.9% (n=20)			
Overextended 8.5% (n=9)			
Disengaged 1.9% (n=2)			
Ineffective & Overextended 7.5% (n=8)			
Overextended & Disengaged 2.8% (n=3)			
Ineffective & Disengaged 0.9% (n=1)			
Burnout 14.2% (n=15)			
100% (n=106)		